UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

Ensuring the Timely and Orderly
Interconnection of Large Loads

Docket No. RM26-4-000

REPLY COMMENTS OF EMERALD AI, INC.

I. INTRODUCTION

Emerald AI respectfully submits these Reply Comments in response to the Notice Inviting Comments issued to the Federal Energy Regulatory Commission's ("Commission") October 27, 2025 Notice Inviting Comments regarding the interconnection of large loads. Emerald AI submits these comments to supplement our initial filing, address the extensive record developed by diverse stakeholders, and to strongly support rewarding verifiable, software-driven flexibility with expedited interconnection.

We align ourselves with the broad coalition of commenters—including the Advanced Energy Management Alliance (AEMA)[1], utilities, transmission owners, and technology companies—in supporting a **Flexible Load Fast Track (FLFT)**. This FLFT would assess load flexibility in a technology-neutral and performance-based manner, rewarding flexibility from a range of different technologies with larger and faster power connections. In addition, Emerald AI writes separately to emphasize that software-defined flexibility of data centers should be eligible as one of many qualifying technologies to provide load flexibility to the power system.

The record demonstrates a tension between two visions for interconnecting large loads. One vision, advocated by a small minority of commenters, views large loads as static liabilities that must "bring their own generation" to neutralize their impact. The second vision, supported by the majority of the record, recognizes that modern AI data centers are dynamic, flexible assets.

Emerald AI provides the "operating system" that makes the second vision possible. Our platform harnesses computational flexibility (both temporally and spatially) and co-orchestrates that compute flexibility with on-site resources like batteries, generators, and cooling systems. Because software-defined flexibility can render data centers responsive to grid conditions without the need for massive, capital-intensive generation buildouts, we strongly push back against any precondition that equates expedited interconnection solely with matching generation.

A broad coalition of stakeholders supports a technology-neutral approach that recognizes software-defined flexibility as a legitimate, credible, and valuable grid asset. The fastest way to

[1] Advanced Energy Management Alliance, Comments on Interconnection of Large Loads to the Interstate Transmission System, Docket No. RM26-4-000 (filed Nov 21, 2025). https://elibrary.ferc.gov/eLibrary/docinfo?accession_number=20251121-5374

connect new loads that advances affordability, rapidly drives investment into U.S. competitiveness and AI infrastructure, and bolsters grid reliability is to harness the full suite of flexibility technologies in a technology-neutral and performance-based way.[2]

II. INDUSTRY CONSENSUS FOR FLEXIBILITY AS A "BRIDGE" TO INTERCONNECTION

The record reflects a convergence of opinion that flexible load capabilities should be leveraged as a "bridge" to interconnection. Crucially, transmission owners and industry bodies have validated that this flexibility need not come solely from physical hardware, but can also come from advanced software orchestration. Emerald AI's software solution is uniquely positioned to build this bridge by converting static load into a dynamic resource immediately, without the multi-year permitting delays associated with building backup power plants.

- Direct Utility Validation: In a significant endorsement of software-defined flexibility, **National Grid** provides direct transmission owner support for this approach, stating they are "actively working to implement solutions to make this possible, including through our engagements with companies such as Emerald AI which focus on advanced analytics and flexible demand management" (page 2). National Grid affirms that large loads, "if integrated with flexibility... can help contribute to overall cost efficiency" (page 2).

- **Google** similarly recommends that the Commission "prioritize reforms that will expedite the interconnection of facilities that maximize the use of existing infrastructure," specifically for loads that "agree to be curtailable" (page 11).

- **AEMA**, in its initial comments, identifies that flexibility includes *"Temporal flexibility... Spatial flexibility... [and] Resource flexibility,"* and it further explains that by utilizing software to shave the peaks of AI data center demand through computational flexibility and on-site energy resources, or shift that computational demand spatially, grid operators can defer these heavy infrastructure investments (page 4).

- **The Industrial Consumer Organizations** also propose the XL Connect framework—"an optional federal, fast-lane interconnection process…to enable rapid, transmission-level interconnection of large electricity loads for customers for whom speed-to-market and/or interconnection certainty is critical, and who can make adequate financial and contractual assurances to the grid" (page 17).

- Various commenters proposed specific mechanisms to operationalize this bridge. **OpenAI** advocates for an "optional curtailable-load pathway" where curtailment obligations allow the transmission provider to "expedite the interconnection request" (page 2).

[2] Luminary Strategies (Arushi Sharma Frank), Comments on Accelerating Speed to Power / Winning the Artificial Intelligence Race RFI, 90 Fed. Reg. 45,032. https://app.halcyon.io/documents/eb341529-2c12-4dd4-baab-f297c6b4e403/attachment

- **Rejecting Hardware Mandates:** The American Conservation Coalition urges the Commission to "maintain technology-neutral rules" and "allow a variety of approaches... rather than prescribing specific solutions." This supports Emerald AI's position that requiring physical generation (as suggested by the PJM IMM, Monitoring Analytics) erodes the value of software-defined curtailment and unnecessarily increases costs.

Emerald AI submits that the Final Rule must be technology-neutral, defining flexibility to include auditable software control systems that deliver performance cited by National Grid, either with or without physical co-location of generation assets. Taken together, these comments confirm the core proposition that underlies Emerald AI's technology and the FLFT: interruptible, verifiably flexible load is not merely a theoretical concept but a practical, near-term bridge that can accelerate large-load interconnections, support AI-driven economic growth, and reduce the need for immediate "steel-in-the-ground" upgrades.

III. THE DEFINITION OF FLEXIBILITY MUST BE TECHNOLOGY-NEUTRAL AND PERFORMANCE-BASED

To operationalize Principle 7 of the ANOPR, the Commission must adopt a definition of load flexibility that is strictly technology-neutral and performance-based. The Commission should reject any definition that conflates "reliability" with "physical generation."

If a facility can verifiably reduce its net load at the Point of Interconnection (POI) within a specified timeframe, the *method* used to achieve that reduction should be irrelevant to the transmission provider. Emerald AI specifically urges the Commission to recognize Software-Defined Flexibility as a valid form of dispatchability and curtailment, equivalent to generation.

The comments filed in this proceeding overwhelmingly support a definition of flexibility that is performance-based rather than asset-specific. A performance-based construct ensures that the Commission adopts a technology-neutral approach, empowering an all-of-the-above portfolio of solutions—software, storage, thermal resources, on-site generation, and other controllable assets—so long as they can verifiably deliver specified flexibility attributes.

Emerald AI has already demonstrated through recent peer-reviewed research (Colangelo et al., *Nature Energy*, 2025) [3] that AI data centers can provide grid-interactive value reliably. Emerald AI's platform harnesses software to deliver load flexibility in three ways:

1. **Temporal Computational Flexibility:** Software intelligently pauses or reschedules non-time-sensitive AI workloads (such as model fine-tuning) instantly upon receipt of a grid signal.

[3] Colangelo et al., "AI Data Centres As Grid-Interactive Assets," *Nature Energy*, 2025. https://www.nature.com/articles/s41560-025-01927-1

2. **Spatial Computational Flexibility:** Software shifts workloads to alternate facilities in unconstrained regions via high-speed fiber, effectively acting as a "virtual transmission line."

3. **Resource Flexibility:** The software co-orchestrates the dispatch of on-site physical assets—such as Battery Energy Storage Systems (BESS), generation, and thermal/cooling management—to deepen the response alongside computational flexibility.

Fluence underscores how BESS can make large loads curtailable and flexible, minimizing the need for expensive and time-consuming network upgrades. The **Thermal Battery Alliance** shows that hyper-flexible thermal loads can avoid periods of peak scarcity while absorbing energy during low-value hours. Taken together with Emerald AI's own demonstrations, the peer-reviewed work in *Nature Energy* on grid-interactive AI data centers shows that these facilities can provide flexible power consumption and relieve grid stress in real time, delivering credible, enforceable, and verifiable curtailment.[4]

IV. THE COMMISSION MUST REJECT PROPOSALS THAT IGNORE DEMAND FLEXIBILITY OR REQUIRE GENERATION PAIRINGS

Emerald AI respectfully urges the Commission to reject arguments in the record that either (1) dismiss demand-side flexibility as "illusory" or (2) require large loads to "bring their own generation" as a precondition for timely interconnection. These positions are inconsistent with the technical evidence in the record and would foreclose cost-effective, software-defined solutions that can provide reliability services equivalent to additional supply.

As Emerald AI explained in its initial comments, the traditional "connect and forget" approach—under which large loads are studied at maximum theoretical withdrawal and assumed to be inflexible—is incompatible with the physical characteristics of modern AI data centers. These facilities are inherently virtual, interconnected, and controllable through software. When paired with an intelligent flexibility platform and enforceable operating envelopes, they can reduce or shift hundreds of megawatts of demand in response to grid conditions, acting as a non-wires alternative to some network upgrades.

Mandating that data centers build matching generation to receive expedited interconnection is economically inefficient and technologically regressive. It ignores the core innovation of the AI era: **we can now move the demand to match the supply, rather than building supply to match static demand.**

- **Cost & Speed:** Building new generation is capital-intensive and delay projects by several years. Software-defined flexibility can be deployed immediately.

[4] Colangelo et al., "AI Data Centres As Grid-Interactive Assets," *Nature Energy*, 2025. https://www.nature.com/articles/s41560-025-01927-1

- **Asset Utilization:** Forcing generation buildout results in redundant assets. Software-defined flexibility maximizes the utilization of *existing* grid infrastructure by shaving peaks and filling valleys.

- **Equivalency:** A reduction in demand (megawatts off) is physically equivalent to an increase in supply (megawatts on) for the purposes of balancing the system.

We agree with AEMA's rebuttal that matching load with generation should be *"just one of multiple pathways... not a mandatory precondition."* The Commission must ensure that software-defined flexibility is treated equivalently and on a technology-neutral basis compared with all other forms of flexibility, dispatchability, and generation.

V. CONCLUSION

The record established in this proceeding demonstrates widespread industry agreement that flexible load is essential to resolving interconnection backlogs. Stakeholders from across multiple sectors—spanning National Grid, Constellation, Google, and Microsoft—have identified that expediting interconnection for flexible resources reduces the need for immediate physical infrastructure and aligns with efficient grid planning.

The path to "Ensuring the Timely and Orderly Interconnection of Large Loads" lies in unlocking the inherent flexibility of the loads themselves. Emerald AI's software platform makes AI data centers flexible assets that bolster grid reliability.

Accordingly, Emerald AI respectfully requests that the Commission establish a Flexible Load Fast Track (FLFT) and adopt a technology-neutral and performance-based definition of flexibility in its proposed large load interconnection rules. In particular, we urge the Commission to explicitly recognize software-defined flexibility as a valid, performance-based resource for expedited interconnection.

Respectfully submitted,
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